CAS3120 Programming Assignment

Overview

- **Task: Tri-class image segmentation**
  - Image segmenation is a pixel-wise classification where a model predicts which class the input pixel belongs to

- **Objective**: Designing convolutional neural network(CNN) based model and understanding Transfer Learning

- **Main Target**: To train the CNN-based model with a given dataset and test it on both in-domain and Out-of-Domain(OOD) data

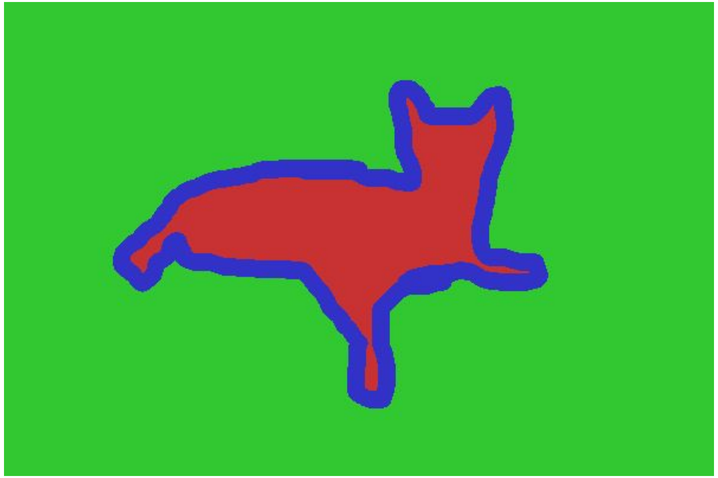
Training Data: Oxford-IIIT pet dataset(CVPR 2012)

- 37 classes of dogs and cats, each class has roughly 200 images(Total: 7349)
  - Species and breed names
  - Bounding Boxes
  - Segmentation masks
- Segmentation masks with three labels(Foreground, Background, Boundary)

- trainval set: 3680, test set: 3669

Training Data: Oxford-IIIT pet dataset(CVPR 2012)

Abyssinian_1.png



Each pixel are labelled as three classes:
1 ➔ Foreground, 2 ➔ Background 3 ➔ Boundary

Implementation

- You have to implement UNet model and training objective function on **main.py** by yourself.
  - The UNet model should follow the original structure of [Ronneberger et al.]
  - Refer to https://github.com/milesial/pytorch-unet
  - Design your own objective function for better performance
  - You can set any of training specifications(Learning rate, batch size, etc.).

- We give you a Dataloader that downloads and loads the training data
  - Check **data.py**
  - You can modify a data transform and augmentation scheme for better score

Kaggle competition

https://www.kaggle.com/t/4b58a142bd494c26bbd073452cf5b656

Please join our Kaggle competition with the above link

- Change your team name to your student ID on the Team Tab
- You can download Oxford dataset(train_images.zip, train_annotations.zip), test_images.zip(for submission), and sample_submission.csv files here
- Check **make_submission.py** to make your submission file.
- Details about rule and submission are provided in our kaggle competition.

Rules

1. **15 submissions** are allowed for the competition
   - **-0.5 points** for extra one submission
   - you can submit 5 submission per day
2. Architecture of the model is limited to **U-Net** family only
3. ImageNet classification pretrained(ResNet, VGG, …) weights are allowed
4. Models must be trainable on a single **NVIDIA RTX 3090**
5. Open-source code and any AI assistants are allowed

Rules

The following are not allowed:

- Segmentation foundation models
- Vision-language pretrained encoders
- Any pretrained weights other than ImageNet classification weights
- External segmentation datasets
- External pet/animal datasets beyond Oxford-IIIT Pet
- Test-time use of any pretrained segmentation model

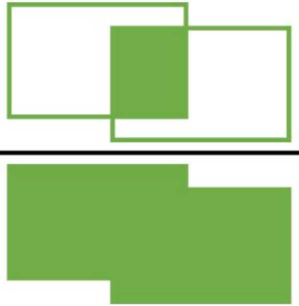
If any violation is caught, you will get 0 point for PA.

If you are unsure whether your action would violate the rule, **please email the TA**.

Evaluation

- Your submissions will be evaluated with mean intersection over union(**mIoU**) across 3 classes(foreground, background, boundary)
- Test set: **244 held-out images** including both in-distribution and out-of-distribution animal categories
- You can check your mIoU score immediately via **Public leaderboard**
- Private leaderboard score is revealed after the competition ends
- Ranking will be scored with Private leaderboard

$$\mathrm{IoU} = \frac{\textit{Area of Overlap}}{\textit{Area of Union}}$$

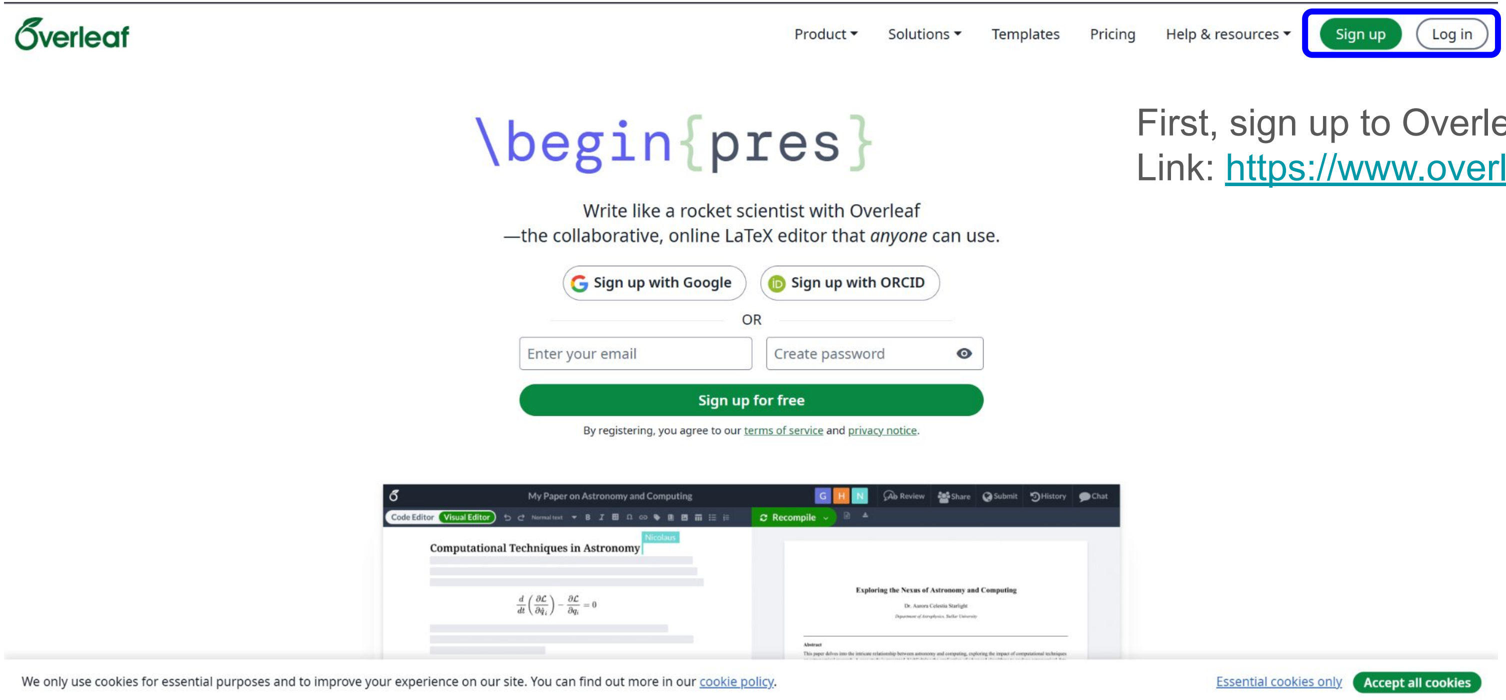


Rules for report

- Write your report on “Example Project” of Overleaf
  - Please read following slides for how to start the example project
  - The report should include introduction, proposed method, experimental results, and concluding remarks section
  - **You must include a reference of the cited papers.**

- The maximum length 5 pages, excluding references

How to make the example project

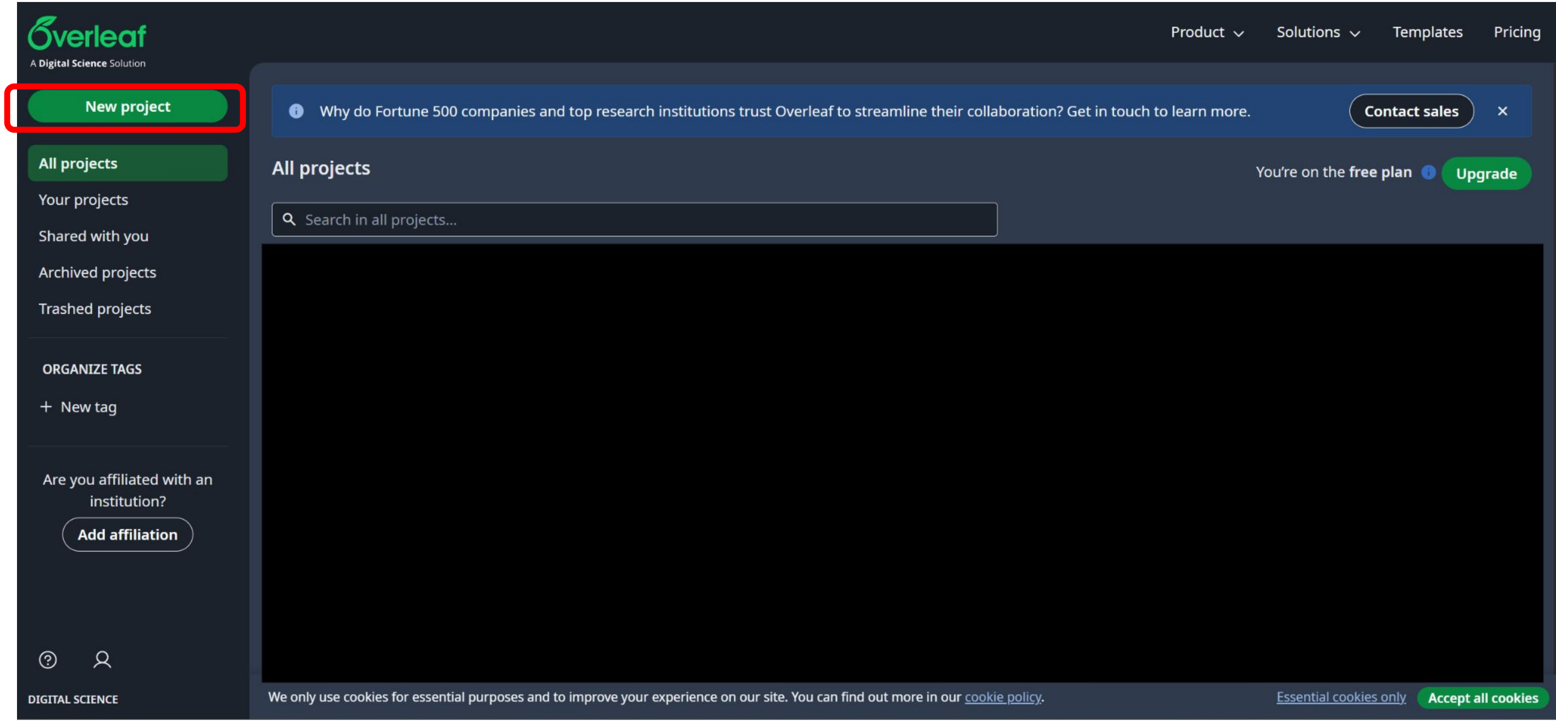


First, sign up to Overleaf

Link: https://www.overleaf.com/

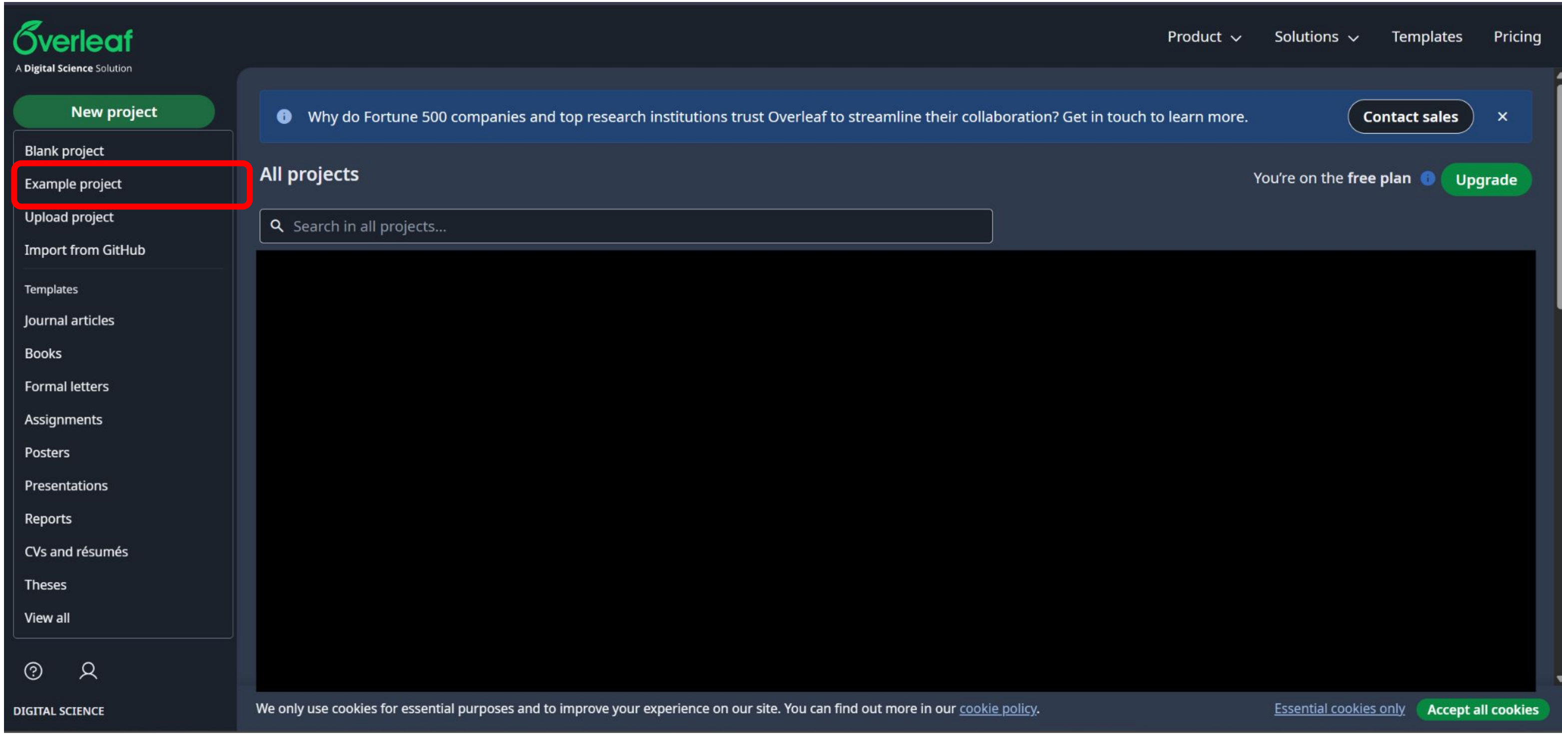
How to make the example project

Click “New Project

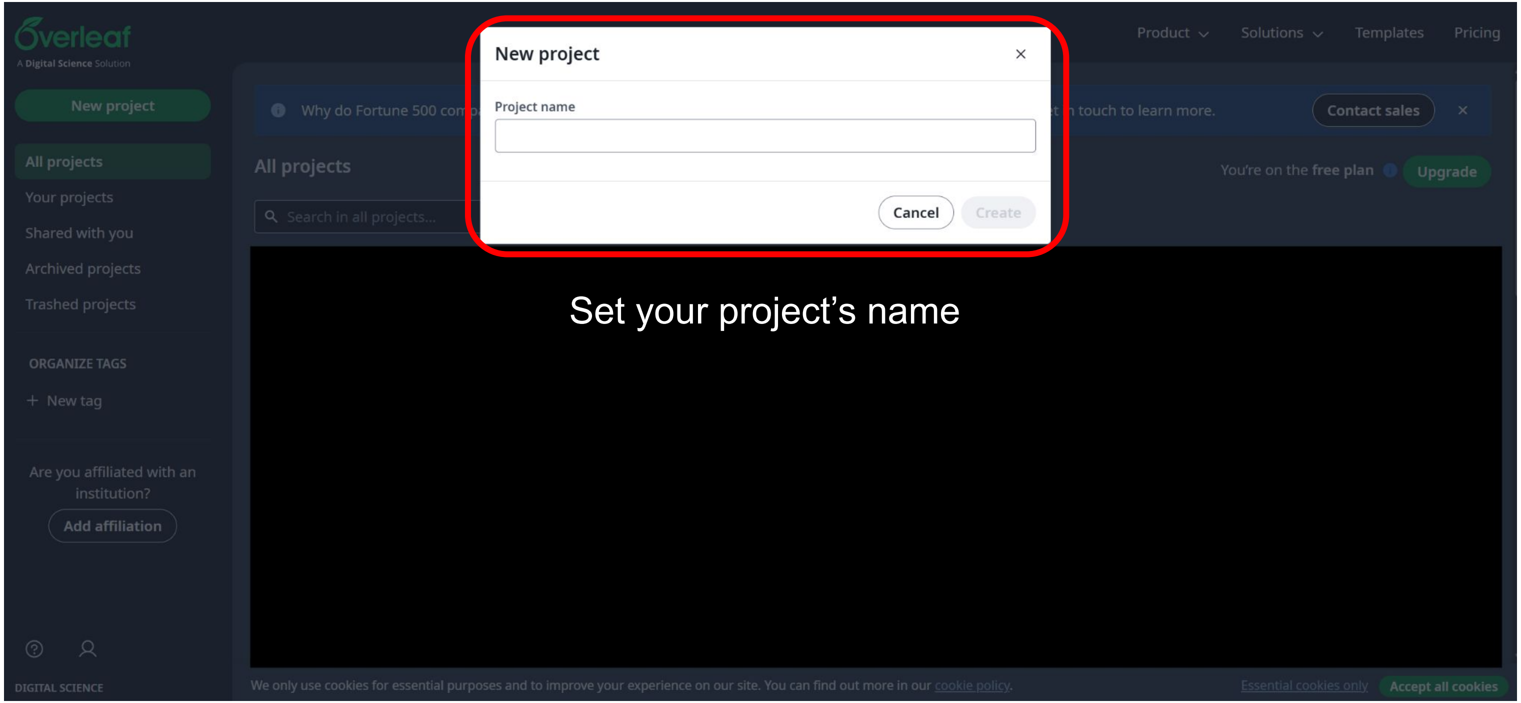


How to make the example project

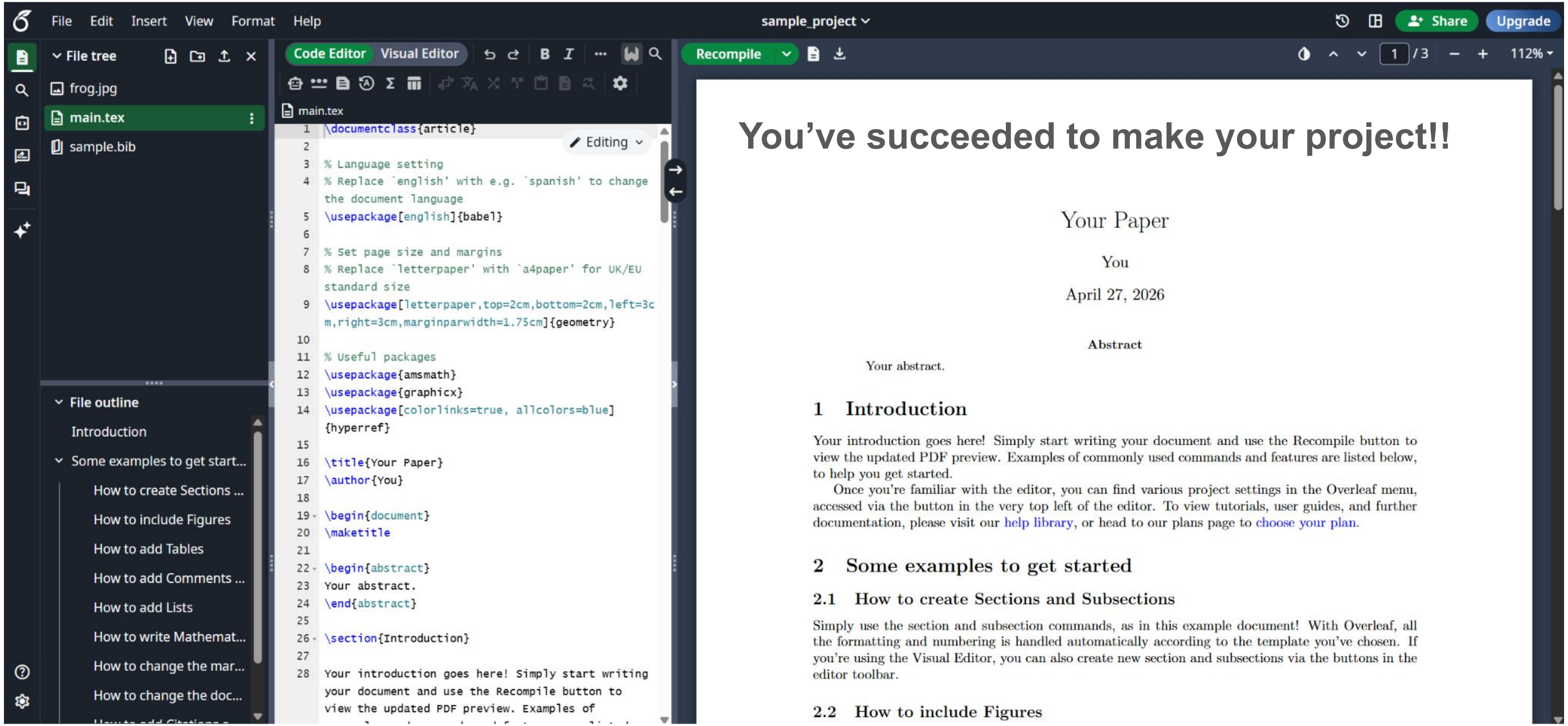
Click “Example Project



How to make the example project



How to make the example project



Rules for report

- Read a content of the example project carefully, you can find out how to write the section and reference
  - Or, you can visit official documentation(Link: https://docs.overleaf.com/)

- You have to write your report by editing the example project, **no other format is allowed**

Submission on LearnUS

- Submit the zip file in LearnUS. The file must have the structure below.
    - [Student ID]_PA.zip e.g. 2024123123_PA.zip
        - Your code files (in.py)
        - report.pdf
        - [Student ID]_submission.csv(submit the csv file which you get the best score)

Grading Scheme(total 20 points)

- Coding(15 points)
  - mIoU > 0.7: 5 points
  - mIoU > 0.8: 10 points
  - Private Leaderboard
    - Ranking #1 - #10: additional 5 points
    - Ranking #10 - #20: additional 4 points
    - Ranking #20 - #30: additional 3 points
    - Ranking #30 - : additional 2 points
- Report(5 points)
  - You can get additional points if you write report well or give a creative solution(Max: 2points)

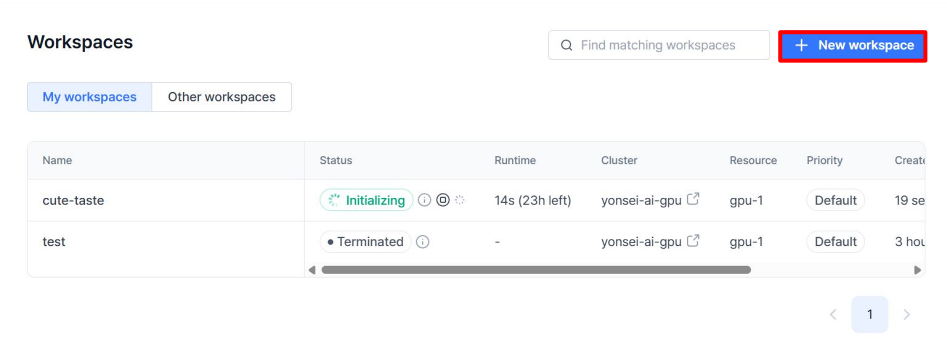
Wrong submission format: -2 points

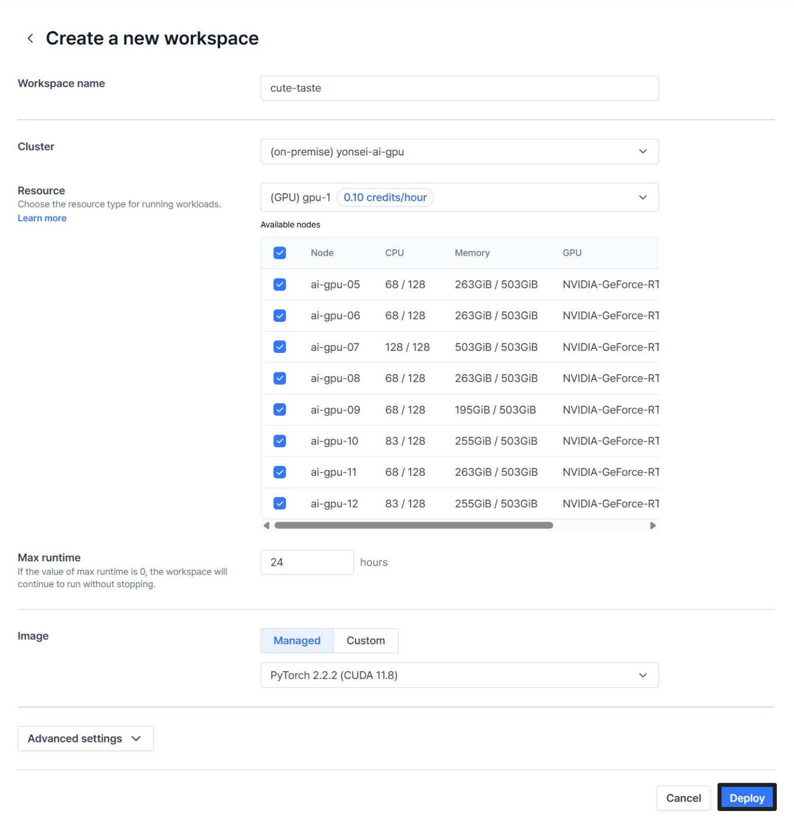
Cut-off score of mIoU and additional ranking points can be changed. If change occurs, we will notice via LearnUS

Due Date

- 31st May, 23:55 KST -> Kaggle submission ends
- 31st May, 23:59 KST -> LearnUS submission ends
- No late submission allowed.

Vessl Setting





Vessl Setting

Caution: you have to turn on admin permission for vscode, git bash, powershell when you install vessl.

python -m pip install --upgrade vessl

Check installation:

vessl --version
vessl configure

Please refer to Yonsei Vessl Manual
We will give you a lecture for setting Vessl